

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

SIAMESE NETS

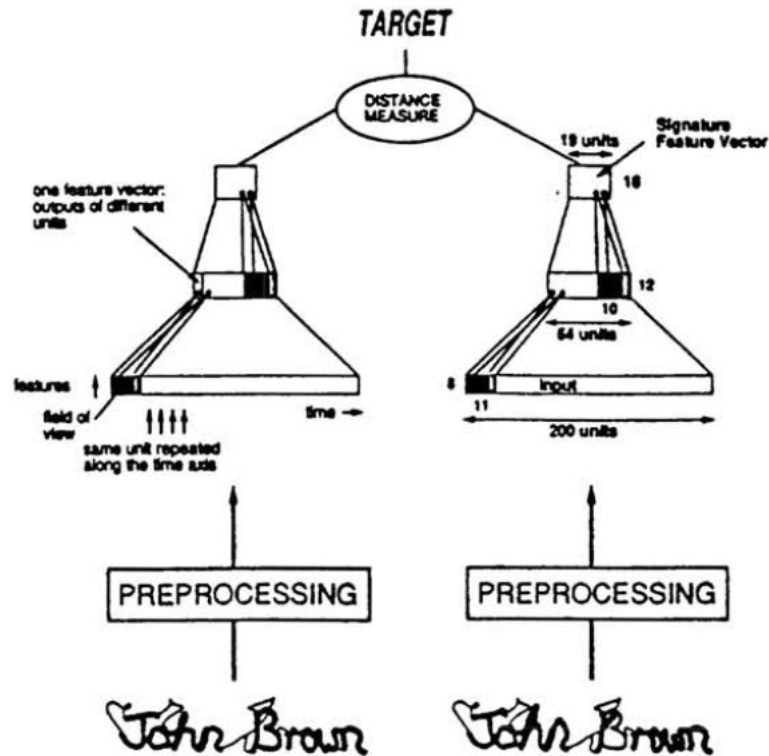
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Application: Checking Signatures for Forgery



Similarity function $s(x, x') = f(\phi(x), \phi(x'))$:

- ϕ is produced by a neural network
- $f(\cdot)$ can be, e.g., cosine similarity
- Net is trained using a loss function that penalizes similar pairs that are far apart and dissimilar pairs that are close together

Bromley, Guyon, LeCun, Sackinger, Shah (1993).

Signature verification using a "Siamese" time delay neural network.